

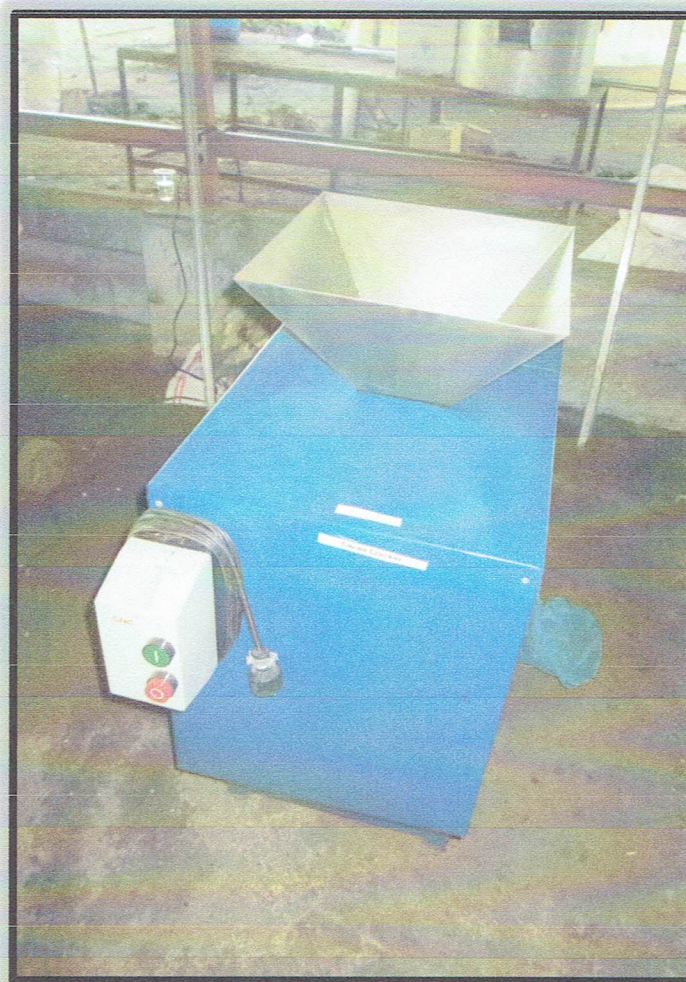


AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2018-0145



Agricultural Machinery – Cacao Huller SHOCK YT38

Test Report Number	2018 – 0145
Valid Until	August 29, 2023
Agricultural Machinery	Cacao huller
Type	Stationary type, hammer type cracker with blower/winnowers
Brand	SHOCK
Model	YT38
Test Requested by	SHOCK & AWE TRADING 301 Lions Building, Candelaria St., Ecoland, Bucana, Talomo, Davao City

SUMMARY

Performance Criteria	Requirements as per PNS/BAFS PAES 253:2018	AMTEC Test
Input capacity, kg/h, minimum	Not specified	53.66
Hulling capacity, kg/h, minimum	Not specified	49.50
Hulling efficiency, %, minimum	98	99.81
Cacao nibs recovery index, minimum	0.85	0.803
Purity, %, minimum	97	98.31

OTHER OBSERVATIONS

1. Manufacturer	SHOCK & AWE TRADING, 301 Lions Building, Candelaria St., Ecoland, Bucana, Talomo, Davao City
2. Prime mover	1.1 kW YL90S/4 Electric Motor
3. Operator's manual	None
4. Previous valid test reports of similar brand and model	None
5. Transporting the machine	Not mobile
6. Cleaning the machine	Accessible
7. Adjusting and repairing parts	Not observed
8. Ease of loading input and collecting output	Hopper height from the ground is 1000 mm. Outlet height from the ground is 220 mm.
9. Safety	Provided with on/off switch and body cover
10. Labor Requirements	Two
11. Failure or abnormalities	Not observed
12. Others	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	April 23, 2018
Location of Test	Ecoland, Bucana, Talomo, Davao City
Items Tested	Operator's Manual; Input and Hulling Capacity; Noise Level; Fuel Consumption; and Product Analysis

METHOD OF TEST

The machine was tested based on the "PNS/BAFS PAES 254:2018 Agricultural Machinery – Cacao Huller – Methods of Test".

DESCRIPTION OF THE MACHINE

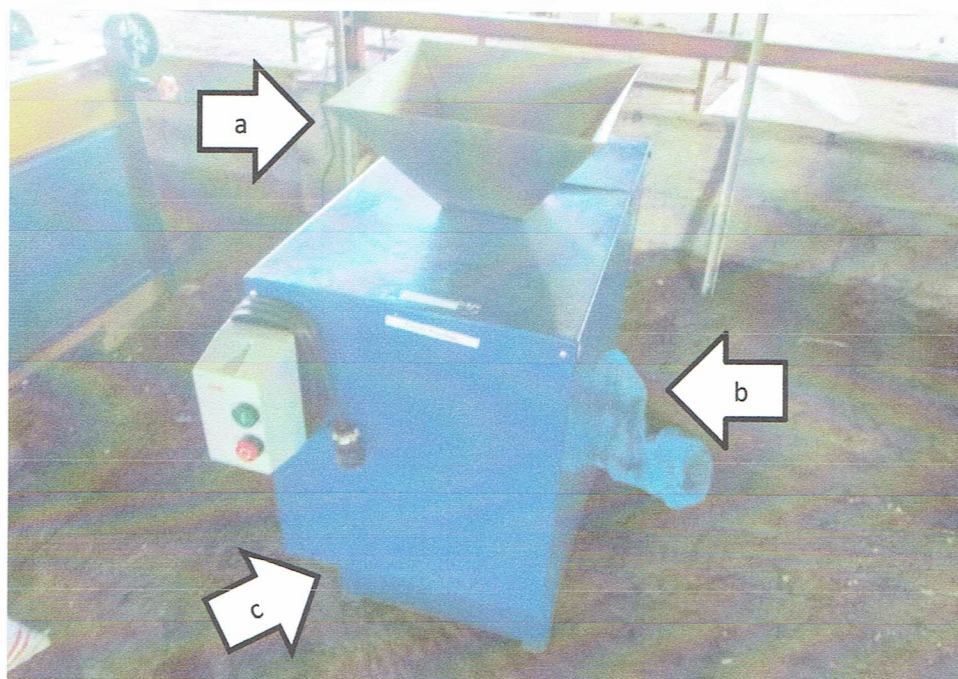


Figure 1. a) Intake hopper
b) blower outlet
c) Output chute

SPECIFICATIONS

Item	Manufacturer's Specification	Verification by AMTEC
1 Main structure		
1.1 Overall dimensions (mm)		
1.1.1 Length	ND	865
1.1.2 Width	ND	410
1.1.3 Height	ND	1000
1.2 Weight, without the prime mover (kg), if applicable	ND	NM
2 Input capacity (kg/h)	120	53.66*
3 Electric motor		
3.1 Brand	ND	ND
3.2 Model	YL90S/4	YL90S/4
3.3 Rated power (kW)	1.1	0.989*
3.4 Rated speed (rpm)	1730	1794*
3.5 Rated voltage (V)	220	227*
3.6 Rated frequency (Hz)	60	NM
3.7 Rated Current (A)	6.1	4.37*
3.8 Power transmission system	ND	Direct coupling
4 Hopper		
4.1 Inlet dimensions (mm)		
4.1.1 Length	ND	70
4.1.2 Width	ND	70
4.2 Height from the ground (mm)	ND	1000
4.3 Material	ND	Stainless steel (1.2 mm)
4.4 Location	ND	Top of the hulling chamber
4.5 Means of attachment	ND	Welded
5 Hulling Mechanism		
5.1 Type	ND	Steel huller
5.2 Dimensions, L × D (mm)	ND	125 × 50
5.3 Material	ND	Stainless steel
5.4 Teeth	ND	
5.4.1 Type	ND	Pegteeth
5.4.2 No. of teeth	ND	4
5.4.3 Material	ND	Stainless steel rod (3 mm)

Note: *Actual performance during operation, NM = Not measured, ND = No data

SPECIFICATIONS (cont'n)

Item	Manufacturer's Specification	Verification by AMTEC
6 Outlet		
6.1 Diameter (mm)	ND	63.5
6.2 Material	ND	BI pipe
6.3 Height from the ground	ND	220
7 Safety devices	ND	On/off switch, body cover
8 Special features	Winnower	Suction type blower
9 Illustration of transmission system	ND	See Figure 2

Note: * Actual performance during operation, NM = Not measured, ND = No data

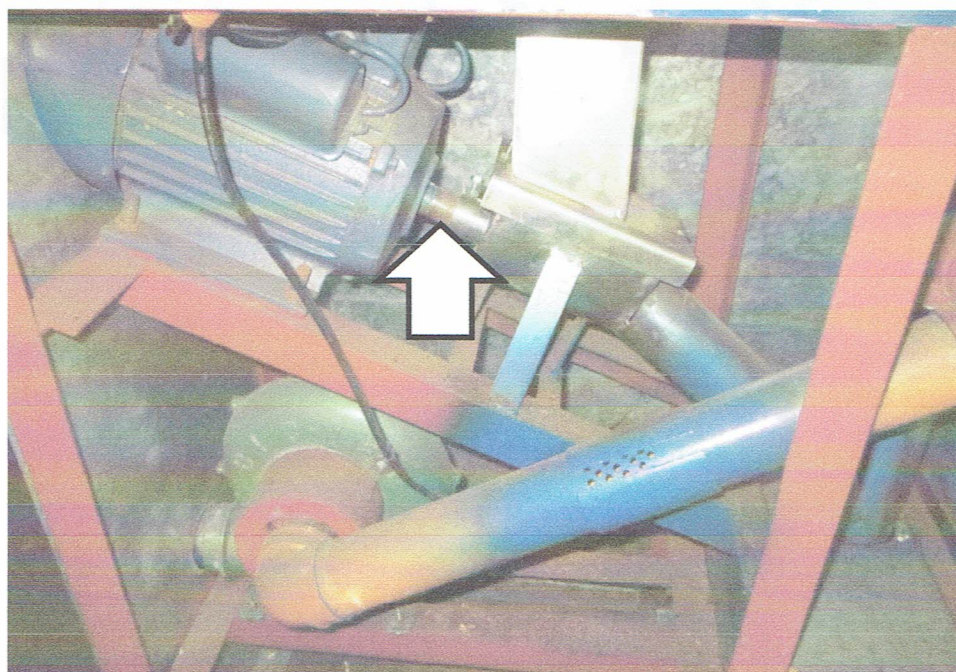


Figure 2. Hulling shaft directly coupled to the electric motor

RESULTS

A. PERFORMANCE TEST

ITEMS	DATA
1 Conditions of dried cacao	
1.1 Source	Davao City
1.2 Variety	ND
1.3 Bean dimension (L × W × Thickness) (mm)	ND
1.4 Coffee bean bulk density (kg/m³)	ND
2 Weight of input cacao (kg)	8.88
3 Input time (h)	-
4 Input capacity (kg/h)	53.66
5 Operating time (h)	0.179
6 Hulling capacity (kg/h)	49.50
7 Weight of output green coffee beans (kg)	7.13
8 Speed of cylinder shaft (rpm)	
8.1 Without load	1797
8.2 With load	1794
9 Noise level [dB (A)]	
9.1 Feeder	
9.1.1 Without load	77.88
9.1.2 With load	76.37
9.2 Bagger	
9.2.1 Without load	78.15
9.2.2 With load	77.31

Note: ND = No data

ITEMS	DATA
10 Power consumption	
10.1 Input power (kW)	
10.1.1 Without load	0.919
10.1.2 With load	0.989
10.2 Line voltage (V)	
10.2.1 Without load	225
10.2.2 With load	227
10.3 Current (A)	
10.3.1 Without load	4.08
10.3.2 With load	4.37

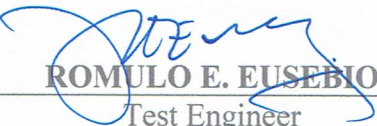
B. LABORATORY ANALYSIS

ITEMS	DATA
1 Conditions of dried parchment coffee	
1.1 Moisture content (% wet basis)	3.01
1.2 Blower loss (%)	0
2 Product analysis	
2.1 Hulling efficiency (%)	80.28
2.2 Hulling recovery (%)	99.81
3 Purity (%)	98.31

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APR 12 2019

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